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Bayesian Optimization with Active Constraint Learning for Advanced Manufacturing Process Design

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ABSTRACT

This study addresses the complex challenge of identifying process parameters for optimal manufacturing outcomes in advanced manufacturing, where nonlinear and costly process-to-quality relationships prevail. We introduce a novel experimental design framework that energizes the optimization of process parameters and feasibility constraint learning with a significantly reduced number of trials as compared to traditional Design of Experiments methods. Our approach is grounded in two primary methodologies: (1) active multi-criteria sample for constraint estimation and (2) Bayesian optimization-based sample for optimal parameter identification. This integration facilitates the efficient discovery of globally optimal parameter settings and outperforms multiple benchmark models in constraint estimation accuracy. The framework's efficacy is demonstrated through application on both synthetic datasets and a real-world case study involving the synthesis of 2D materials, demonstrating its potential to enhance manufacturing efficiency and quality in complex manufacturing processes significantly.

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1. Introduction

In the realm of manufacturing, the optimization of process parameters is crucial to achieving the objectives of enhanced productivity and quality. The advent of advanced manufacturing technologies introduces complex process dynamics, wherein the relationship between process parameters and quality is often nonlinear and nonstationary. This complexity presents a significant challenge, making the analysis tedious and time-consuming. This complexity necessitates a deeper exploration to unravel the relationship between the process parameters and the process performance. Processes such as material synthesis and additive manufacturing are two examples of such complexities. Identifying the optimal process parameters with minimal experimental effort is particularly challenging, especially when manufacturing costs limit the number of feasible experimental trials. Therefore, it's desirable to develop a fast yet accurate method to identify optimal process parameters efficiently using the fewest number of trials.

The conventional approach to process configuration and optimization in manufacturing involves traditional Design of Experiment (DoE) methods based on statistical principles, such as Factorial Designs, Response Surface Methodology (RSM), and Latin Hypercube Design (LHD). Optimal candidate combinations of process parameters are proposed based

on regression analysis. Alternatively, data-driven modeling using supervised machine learning models has been employed to capture the process-to-quality relationships. However, these two passive learning methods require a large number of samples to achieve a good predictive performance due to the high complexity of the manufacturing system. Sequential design methods have become increasingly prevalent in engineering applications to address the significant challenge of large-scale human labeling efforts, which are both expensive and time-consuming. One of the key advantages of employing data-driven sequential design methods is their ability to extrapolate insights from a small initial dataset to reduce the high cost of the labeling process. Active learning (AL) and Bayesian Optimization (BayesOpt) are two subfields of machine learning-based sequential design methods that iteratively query the most informative unlabeled data via acquisition function to refine the surrogate model of advanced manufacturing processes with a limited amount of data (Bukkapatnam, 2023). It is worth highlighting that the main difference between AL and BayesOpt is the choice of acquisition function. Acquisition functions used in AL prioritize selecting unlabeled data from regions characterized by high uncertainties. This strategy aims to improve the overall accuracy of the surrogate model. By contrast, acquisition functions in BayesOpt focus

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more on finding the global optimal of the objective function without considering the reliability of the surrogate model.

Recent advances in BayesOpt in advanced manufacturing include but are not limited to 1) Alternative surrogate model: the DGP-SI-BO method (Gnanasambandam *et al.*, 2025) used a deep Gaussian process as a surrogate to model the non-stationary functions in Laser Powder Bed Fusion (LPBF) process. 2) Multi-objective BayesOpt: which has the goal of optimizing multiple properties using an iterative Pareto optimal design process (Jung *et al.*, 2023). Multi-task GPs (Shen *et al.*, 2023) are selected to learn multiple outputs simultaneously for correlated multiple properties. 3) Collaborative BayesOpt: multiple designs are chosen from a common surrogate and to be evaluated parallel via different agents (Yue *et al.*, 2023). BayesOpt (sequential design methods) is prominent in manufacturing applications, yet they consistently need to account for constraints in process parameters. Generally, a constrained optimization problem is usually formulated to achieve a sequential design in engineering applications where parameter constraints are already known (Guidetti *et al.*, 2021). Additionally, in the mechanical system design process, it's essential to consider constraints related to the process outcome (e.g., desired/undesired quality of the outputs). However, these constraints are often unknown in the design space and can only be determined after experimental evaluation. Checking if a design satisfies a constraint often requires computational modeling or resource-intensive experiments (Lee *et al.*, 2023).

Existing research on sequential design under unknown constraints (Gelbart *et al.*, 2014) has demonstrated its ability to optimize the objective function. However, the iterative nature of this constrained BayesOpt algorithm does not facilitate learning about feasibility. The constrained boundary distinguishing “can grow” and “cannot grow” design space in material science is crucially informative (Costine *et al.*, 2020; Khatamsaz *et al.*, 2022). Therefore, developing a flexible sequential design method that simultaneously incorporates feasibility constraint learning and optimizing the target process holds great promise for enhancing advanced manufacturing processes. By accurately identifying the feasible design space, optimization can be more effectively confined to this area, significantly increasing the efficiency of the design process. Nonetheless, the label scarcity issue in manufacturing experiments limits the model's capability in the loop, which poses a significant challenge to handling constraints and ensuring the feasibility of the design.

To address the key challenge mentioned in BayesOpt with incomplete knowledge of constraints, previous research (Khatamsaz *et al.*, 2023, 2022) proposed a framework that synergizes AL and BayesOpt to actively learn the undefined constrained boundary and optimize the process parallel. A Bayesian classifier is adopted to learn the feasibility region, and then unlabeled data with the highest prediction uncertainty measured by entropy is queried in each iteration. However, the scarcity of labels and the complexity of the nonlinear relationship of process hinder the

performance of active learning in constraint identification, which may lead to inaccurately classifying the true global optimal sample as part of infeasible regions. For example, Figure 1(a) shows the ground truth of the objective function where the global optimum $f^*(x) = f(0, 0)$ is located in the center of the whole design space, and there are four infeasible regions.

Figure 1(c) shows the initial prediction result of the constraint model trained via labeled data shown in Figure 1(b). As shown in Figure 1(c), the global optimum is misclassified as an infeasible sample far from the predicted constraint boundary. Refining the feasible boundary via standard active learning methods like uncertainty-based sampling (Khatamsaz *et al.*, 2023, 2022), as shown in Figure 1(d), would trigger numerous additional human annotation processes to estimate the feasible design space that incorporates optimal design correctly. The limited experimental dataset and ineffective active learning strategy cannot ensure the accuracy of prediction for feasible design space, making the search for optimal design stuck in a local-searching scenario where the true global optimum is ignored.

To accelerate the optimization process and prevent the optimal design from being misclassified as infeasible, we developed a novel sequential experimental design method that employs two collaborative loops to explore the feasibility constraint boundary and optimize the process simultaneously. The proposed method, depicted in Figure 2, consists of two main components that work collaboratively in an iterative manner: (1) optimization loop toward the desired process outcome, and (2) active constraint learning (ACL) loop to refine the feasibility constraint boundary. More specifically, the ACL iteratively re-estimates the feasible region, progressively narrowing it over time to focus on the most promising search area that is likely to contain the global optimal point. To enhance the efficiency of our framework in simultaneously optimizing the target process and learning constraints, we have devised a sequential sampling strategy founded on three key measurements: (1) **representativeness**, (2) **uncertainty**, and (3) **diversity**. This strategy utilizes a unified function within the active loop to select the most informative unlabeled samples. Based on these three metrics, the query generated by the optimization loop will also influence the selections in the ACL loop and each selection in the ACL loop subsequently impacts the next choice. This approach can improve the optimization's convergence speed and facilitate the identification of the feasibility constraints with the least human effort. Our contributions in this paper are as follows.

1. We develop a novel sequential experimental design method that leverages two loops collaboratively to avoid the risk that the optimal design is misclassified as an infeasible design. The designed framework avoids extensive human labeling via more effective constraint learning and faster convergence to the objective.
2. We present a new multi-criteria sampling strategy that merges representativeness, uncertainty, and diversity measures. This strategy enhances the exploration of the

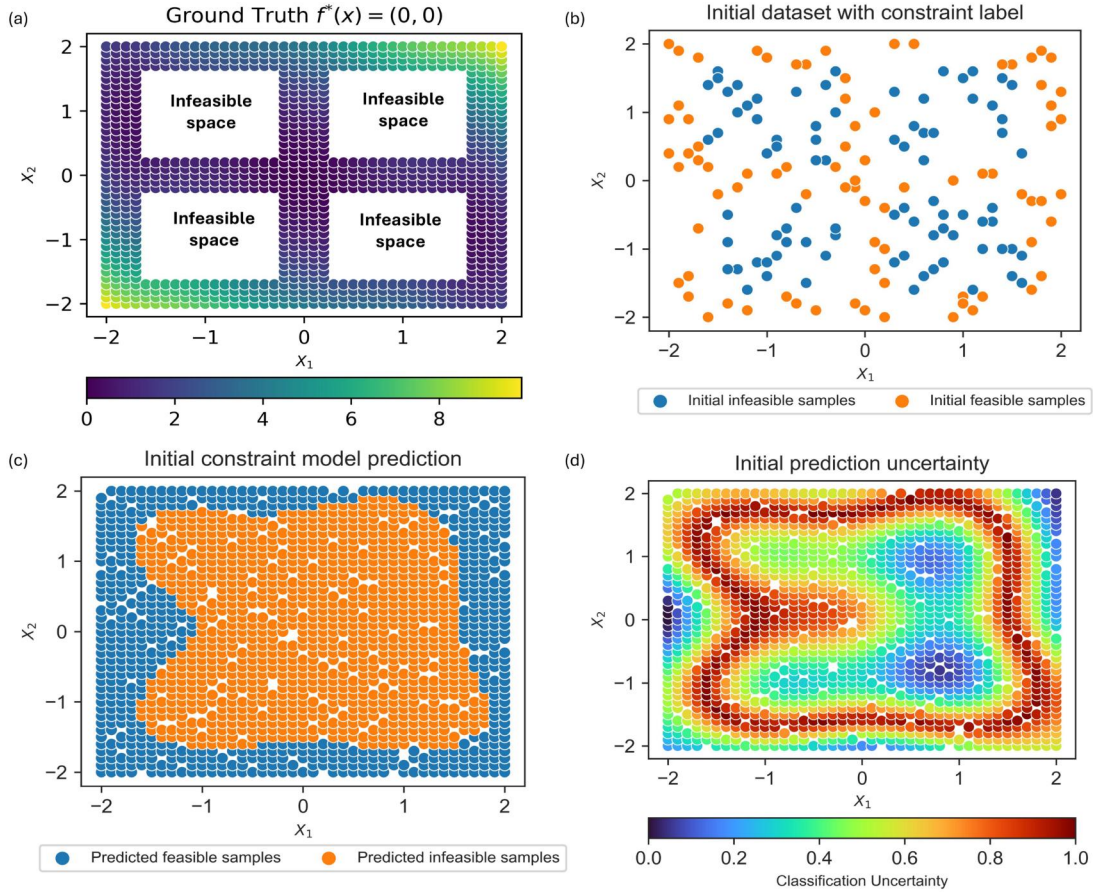
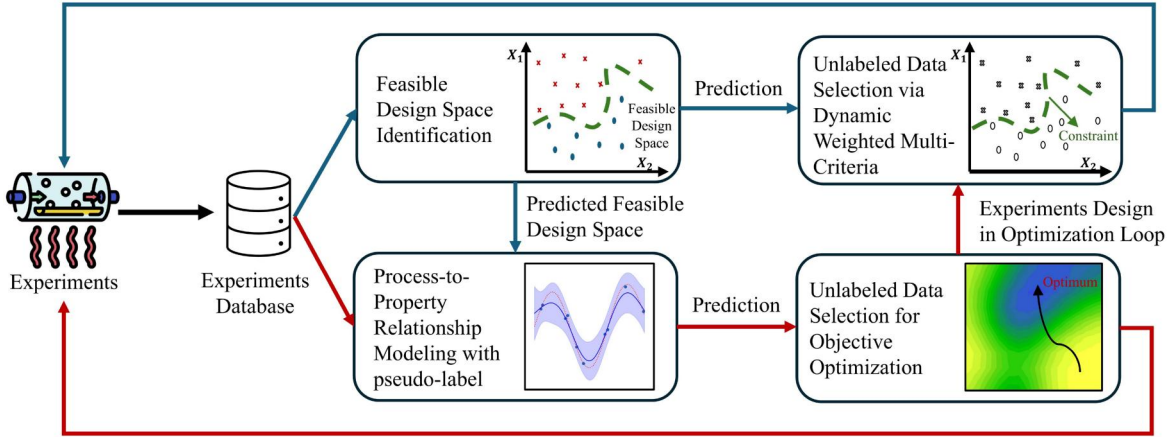


Figure 1. The illustration of the limitation of current works. (a) The ground truth of the objective function and infeasible regions; (b) initial dataset with the label of feasibility; (c) initial constraint model prediction; (d) prediction uncertainty of constraint model.

Algorithm 1: Experiment Design for Active Constraint Learning



Algorithm 2: Experimental Design for Objective Optimization

Figure 2. Overview of the proposed framework.

implicit constraint boundary, enabling more efficient identification of the feasible region within the design space and thereby improving the optimization of the target process.

3. We propose a novel heuristic sampling strategy to quantify the joint informativeness of samples queried in each iteration, further refining the sample selection process and improving optimization efficiency.

The remainder of this article is organized as follows. Section 2 reviews the literature on AL and BayesOpt in manufacturing applications and sequential sampling methods under unknown constraints. Section 3 provides an overview of the proposed framework and details for proposed algorithms. In Section 4, the proposed approaches are evaluated with two case studies: (i) synthetic datasets for framework algorithms analysis and (ii) an advanced

manufacturing case of the 2D material synthesis process. Lastly, a summary of this paper is provided in Section 6.

2. Literature Review

In this section, we discuss related literature on AL and BayesOpt, especially focusing on sequential sampling strategies in the related works. While AL and BayesOpt have different objectives, their integration forms the foundation of our design approach. Our focus is specifically on research that applies AL and BayesOpt within the engineering domain.

2.1. Sampling Criteria in Active Learning

Active learning is widely adopted in machine learning applications when the labeling process is costly. Uncertainty and diversity are two of the most adopted criteria of AL for selecting unlabeled samples. Uncertainty-based methods, such as query-by-committee (Burbidge *et al.*, 2007) and entropy-based sampling (Settles, 2009), choose unlabeled data with the most uncertain prediction to annotate labels for model retraining. Meanwhile, diversity represents the similarity between labeled and unlabeled data. Diversity-based methods choose the most dissimilar samples to query via different measurements, such as Euclidean distance and Cosine similarity. Previous research (Wu *et al.*, 2019) designed a greedy sampling strategy to select the farthest sample from labeled data in the input and output space. Another study (Wang *et al.*, 2017) proposed an AL sampling method that integrates uncertainty and diversity criteria via sparse modeling. However, samples are disproportionately labeled in different classes in active learning for real-world classification applications. Purely uncertainty-based and diversity-based methods will focus more on the majority class, which can degrade the performance. To tackle this problem, representativeness-based sampling strategies were proposed to quantify the impact of data density on the respective classifiers. Previous research (Huang *et al.*, 2010; Du *et al.*, 2015) combined the criteria of uncertainty and representativeness to optimize selection. Similarly, a recent study (Park and Kim, 2019) proposed a cluster-based sampling strategy to unify the diversity and representativeness of unlabeled data to assign queries to different classes.

Studies have demonstrated that sampling methods considering multiple criteria simultaneously can improve AL performance. However, most previous research (Shim *et al.*, 2021; Liu *et al.*, 2021; Park and Kim, 2019) on batch mode multi-criteria AL primarily focus on clustering-based strategies. They usually first applied clustering methods to capture the underlying structure of unlabeled data and then defined representativeness and diversity based on the clustering results. However, identifying accurate clustering boundaries is challenging when unlabeled experimental samples are uniformly distributed across the predefined design space. In such cases, a recent study (Kee *et al.*, 2018) measured representativeness and diversity using k -nearest neighbors and integrated uncertainty through a linear

combination with static and predefined weighted parameters. Thus, we aim to enhance the AL by implementing a dynamic weighted multi-criteria sampling strategy. As a key element of our framework, the proposed active sampling method for the constraint model plays a significant role in improving decision boundary exploration. It also collaborates with the optimization process, thereby increasing the rate of optimization speed.

2.2. Bayesian Optimization under Unknown Constraints

In machine learning literature, there are two main methodologies in BayesOpt under implicit constraints. This first type of previous work (Gardner *et al.*, 2014; Lam and Willcox, 2017) is interested in finding the global minimum x_* of an objective function $f(x)$ subject to non-negativity of a series of constraint functions $c_1, c_2, \dots, c_k$. However, objective function $f(x)$ and constraint functions c_i are unknown and can only be evaluated by experiments. They followed the design idea of Schonlau *et al.* (1998) who proposed a constrained acquisition function with a latent constraint function that multiplies the probability of feasibility for each constraint. Some updated versions of constrained acquisition functions, like the FIA function (Hickman *et al.*, 2023), incorporate a c' term that calculates the ratio of infeasible measurements to the whole design space or user feedback with a higher probability of feasible design (Jetton *et al.*, 2024). These methods refer to a safety-aware sampling strategy that integrates objective functions and constraints to make solutions feasible. However, the multiplier of feasibility probability cannot update the estimated feasible region to make constraints known.

An alternative approach to managing undefined constraints involves integrating an individual sampling criterion for constraint exploration. Recent work (Lee *et al.*, 2023) designs a linear combined acquisition function integrating a sampling strategy of feasible design updation and target process exploration. However, this work focuses on optimizing surrogate models globally without searching for optimal process parameters. In addition, a The STAGEOPT framework (Sui *et al.*, 2018) was proposed to expand the estimated feasible region in the first stage and then implement typical BayesOpt within the predicted feasible space. A parallel optimization framework named pBO-2GP-3B (Tran *et al.*, 2019) combines three acquisition functions to query a batch of unlabeled samples in each iteration. However, previous work (Khatamsaz *et al.*, 2023, 2022) always regards the AL and BayeOpt processes as two individual parts that propose queries independently. Within the AL process, they used a single criterion for the feasible region exploration, which is not very informative. The information extracted from the prediction in the first couple of iterations is unreliable due to the data scarcity problem. Consequently, overcoming the challenges posed by limited experimental data represents a critical research gap in applying Bayesian optimization within the manufacturing domain.

3. Methodology

This paper proposes a novel experimental design method incorporating two feedback loops to address the optimization challenges complicated by unknown feasibility constraints. In our framework, we employ two distinct models functioning collaboratively. The first, a constraint model, is utilized to estimate the boundaries of the feasible area within the design space. The second, a surrogate model, is designed to learn the objective function. These two models are trained simultaneously and operate through two distinct yet interconnected feedback loops—an optimization loop (in red) and an active learning loop (in blue)—as illustrated in Figure 3. The constraint model predicts a feasible region, Ω_f , which helps the objective loop concentrate on identifying an optimal design within a reduced design space (indicated by the downward blue arrow in Figure 3). In this framework, each loop employs its own strategy for querying new data in each iteration. The **optimization loop** first identifies the optimal data point x^* within the predicted feasible region Ω_f , while the **active learning loop** utilizes a multi-criteria active sampling approach incorporating x^* (as shown in the upward arrow in Figure 3) to query new data points X_u for updating the boundary for the feasible region. Together, they collaborate not only to optimize the objective process but also to reveal the relationship between process variables and the resultant properties (outputs vs feasibility). The overall framework is presented in Figure 3. The proposed methods address the following fundamental questions:

- How to devise a unified framework to let two distinct loops work collaboratively?
- What are the optimal criteria for selecting the most informative samples from unlabeled data, enhancing the precision of constraint estimation for subsequent process optimization?
- What is the best strategy for sequential selection to explore the unknown constraint while optimizing the process effectively?

3.1. Problem Statement

Consider a manufacturing process defined over a predefined design space $\mathcal{X} \subseteq \mathbf{R}^D$, where $\mathcal{X}$ is the space of a set of process parameters. The process includes a target function $f: \mathcal{X} \rightarrow \mathbb{R}$, which we want to optimize, and a constraint

function $h: \mathcal{X} \rightarrow \mathcal{Y} = \{0, 1\}$, related to the feasibility of the process which is assumed to be independent of f . For any design point $x = [x_1, \dots, x_D]^T \in \mathcal{X}$, the process will be feasible when

$$h(x) = 0 \quad (1)$$

and infeasible otherwise. For example, f could represent the structure toughness of an additively manufactured part determined by a given vector of design parameters $\mathbf{X}$. Structure toughness is an important property to maximize in the design content for safety and failure tolerance. h can be the boundary to classify the success and failure of structure design given on the same set of design parameters $\mathbf{X}$. To find the optimal process parameters $\mathbf{x}^*$ within feasible space, which can be regarded as a constrained optimization problem in turn

$$\arg \min_{x \in \mathcal{X}} f(x) \quad (2)$$

$$\text{s.t. } h(x) = 0$$

Here, we lack knowledge of the special structure of $f(x)$, like concavity and linearity. We assume $f(x)$ can only be observed or evaluated with costly real experiments or relevant physics-based models as

$$y = f(x) + \epsilon$$

where $\epsilon \sim \mathcal{N}(0, \nu_f^2)$ the observation noise. The $f(x)$ is always summarized as a "black box" (Frazier, 2018). The derivatives of $f(x)$ are not evaluated. We let $h(x)$ model the feasible region $\mathcal{X}_s$, the subset of design space with successful experimental settings, which is also unknown and difficult to obtain. We refer to the complementary of the feasible region as the infeasible region such that $\mathcal{X}_f = \mathcal{X} \setminus \mathcal{X}_s$. Suppose we have n samples from the design space denoted by $\mathcal{D}_n = \{(\mathbf{x}_i, y_i, c_i)\}_{i=1}^n$, and $\hat{f}$ and $\hat{h}$ are our predictors of f and h trained with the current dataset $\mathcal{D}_n$. This work assumes that both f and h follow a Gaussian Process (GP) prior.

$$f \sim \mathcal{GP}(\mu_f(\cdot), \sigma_f(\cdot)), h \sim \mathcal{GP}(\mu_h(\cdot), \sigma_h(\cdot))$$

where $\mu(\cdot)$ is a mean function and $k(\cdot)$ is a covariance kernel function (Rasmussen, 2003). In this work, we assume $f(x)$ and $h(x)$ are independent functions and are modeled as independent GPs. Generally, as long as $f(x)$ and $h(x)$ are modeled independently, they do not need to follow the same follow type, provided both $f(x)$ and $h(x)$ can quantify

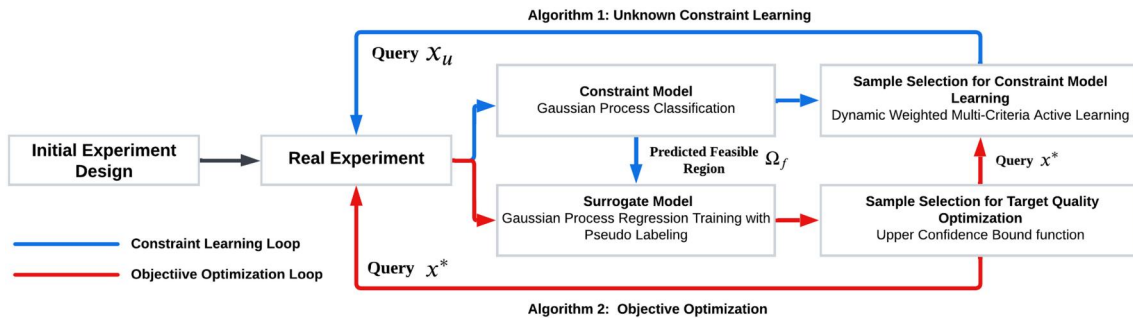


Figure 3. Illustration of the proposed framework workflow.

prediction uncertainty. This also refers to (Gelbart *et al.*, 2014). The models are not restricted to GPs and can also include alternatives such as Bayesian linear regression and Bayesian deep neural networks. We use GPs in this study as they are the most widely used surrogate models in BayesOpt, especially in manufacturing process design where the dimension of the input feature is less than 20.

3.2. Active Learning Loop for Unknown Constraint Learning

There are two parts in the active learning loop (blue line) in Figure 3. A constraint model is adopted to approximate the unknown feasibility constraint, which provides an estimated feasible region varied iteratively for searching the global optimum. In addition, a novel active sampling strategy based on three sampling criteria is designed to refine the constraint learning more effectively. Thereafter, we discuss in detail on a heuristic-based multi-criteria sampling strategy to identify the most informative samples in a batch mode to refine the approximation of the constraint with the least human effort.

3.2.1. Constraint Model in the Active Learning Loop

In this work, we use the GP classification model to learn the constraint boundary of process feasibility. Given a set of observed experiments, denoted as $\mathbf{X} = [x_1, x_2, \dots, x_n]$, and their corresponding feasibility labels $\mathbf{C} = [c_1, c_2, \dots, c_n]$, we have target label feasibility label $C \in \{0, 1\}$ is followed by a Bernoulli distribution. Given observed data $\mathbf{X}$, our goal is to predict target c_* at data point x_* using the predictive distribution $p(c_* = 0|x_*, \mathbf{X}, \mathbf{C})$. In this work, we adopt the Expectation Propagation algorithm (Williams and Rasmussen, 2006) to approximate the posterior $q(h(x)|\mathbf{X}, \mathbf{C})$ which is non-Gaussian via the Gaussian approximation

$$q(h(x)|\mathbf{X}, \mathbf{C}) = \mathcal{N}(h(x)|\hat{\mu}(\cdot), \hat{\mathbf{k}}(\cdot, \cdot)) \quad (3)$$

specified by a mean function by $\hat{\mu}(\cdot)$ and a positive definite covariance function $\hat{\mathbf{k}}(\cdot, \cdot)$. This paper considers the automatic relevance determination using the radial basis function (RBF) kernel function. The distribution of the latent variable $h(x_*)$ corresponding to a new sample x_* is

$$p(h_*|x_*, \mathbf{X}, \mathbf{C}) = \int p(h_*|x_*, \mathbf{X}, \mathbf{h})p(\mathbf{h}|\mathbf{X}, \mathbf{C})d\mathbf{h} \quad (4)$$

where $\mathbf{h} = [h_1, h_2, \dots, h_n]^T$, and then the prediction distribution can be computed via

$$p(c_* = 0|x_*, \mathbf{X}, \mathbf{C}) = \int \Phi(h_*)p(h_*|x_*, \mathbf{X}, \mathbf{C})dh_* \quad (5)$$

Given the predictive probability, we assume the prediction label of x_* is the most probable label

$$\arg \max_{c_*} p(c_*|x_*, \mathbf{X}, \mathbf{C}) \quad (6)$$

In each iteration, the constraint model generates a predicted feasible design space Ω_f for subsequent unlabeled data selection in the optimization loop. This predicted

feasible design space Ω_f provided a small but promising region for global optimal searching, and it will also be updated iteratively as more data is labeled in both loops. In addition, the predictive probability will also be utilized to measure the information of unlabeled data to guide data selection in the constraint learning loop.

3.2.2. Sampling Criteria for Unlabeled Data Selection

In this active constraint learning loop, three criteria are adopted jointly to quantify the information carried off each unlabeled sample. In these three criteria, (1) *Prediction uncertainty* measures the confidence of the current classifier; (2) *Representativeness* reveals the hidden pattern of unlabeled data; and (3) *Diversity* is introduced to maximize the information in the selected batch. It is worth mentioning that in this work, There are two goals of active sampling via multiple sources of information. The first goal is to improve the optimization efficiency, and the secondary goal is to learn the constraint function.

Uncertainty: for unlabeled sample x_i , the confidence of prediction is measured by the difference between the prediction probability of the feasible class and the infeasible class, that is:

$$p(h(x_i) = 0|x_i) - p(h(x_i) = 1|x_i)$$

Larger confidence indicates the classifier has a higher prediction certainty of sample x_i , as equation (6) will dominate the prediction result. Meanwhile, we can also find that samples with larger confidence are located far away from the boundary of the classifier. Therefore, such a sample is less informative and is less likely to be selected. However, in the active sampling process, we are more interested in selecting samples that lie near the decision boundary of two classes, which means such a sample enjoys a higher prediction uncertainty. In this work, we define $S_u(x_i)$, the uncertainty of prediction, as

$$S_u(x_i) = 1 - |p(h(x_i) = 0|x_i) - p(h(x_i) = 1|x_i)| \quad (7)$$

In other words, samples with larger $S_u(x_i)$ are more likely to be selected to refine the decision boundary.

Representativeness: The representativeness represents how much similar information one sample x_i can carry compared with all remaining unlabeled samples. It is defined as:

$$S_r(x_i) = \frac{1}{N_u - 1} \sum_{x_j, x_j \in \mathcal{U}, i \neq j} \|x_i - x_j\| \quad (8)$$

where $S_r(x_i)$ is the average Euclidean distance between the unlabeled sample x_i and all remaining unlabeled samples. The unlabeled sample with the minimum value of $S_r(x_i)$ enjoys the highest representativeness. The representativeness is crucial, especially in the initial stage of the optimization process, where prediction uncertainty is generated from the constraint model is not perfect owing to the scarcity of labeled data. The representativeness is utilized to explore the decision boundary via a geometrical feature of unlabeled data to compensate for the limitation of purely uncertainty-based strategy

Diversity: In batch-mode AL, when a batch of unlabeled data is selected for the next iteration, single or both representativeness and uncertainty-based criteria would still not be sufficient to find a group of "informative" samples. To consider redundant information among samples, the diversity of selected samples is introduced to avoid repeatedly selecting samples with high similarity. Diversity term denoted as follows:

$$S_d(x_i) = \min \|x_i - \mathbf{X}_{m-1,t}\| \quad (9)$$

to measure the dissimilarity between unlabeled samples and selected samples, including labeled samples and selected samples in the current selection batch, $\mathbf{X}_{m,t}$ is a set of selected samples in the iteration t and m is a hyperparameter for batch size. For example, in each iteration t , $X_{0,t}$ is the labeled dataset and $x_{m-1,t}$ is the last selected sample in the current batch.

3.2.3. Heuristic Multi-Criteria Sampling Strategy

Considering all three criteria mentioned in Section 3.2 simultaneously, we proposed a linear aggregated function $Q(x_i)$ to select a batch of samples from the unlabeled dataset, which is defined as follows:

$$Q(x_i) = (1 - \alpha - \beta)S_u(x_i) + \alpha S_r(x_i) + \beta S_d(x_i) \quad (10)$$

where $x_i \in \mathcal{U}_s$, $\alpha, \beta \in [0, 1]$ are two non-negative weighted parameters and $\alpha + \beta = 1$, which control the decision-making preference between each criterion. The criterion of representativeness $S_r(x_i)$ is designed to explore the uncharted design space by quantifying and maximizing the information x_i carries. This is particularly crucial in the initial iterations when only a few experimental data points are observed. The measure $S_u(x_i)$ aids in refining the constraint boundary, which has growing significance for active learning as more labeled data become available. The diversity measure, $S_d(x_i)$, is introduced to minimize redundancy in labeled data and prevent over-concentration of the selected samples with small local regions. Shifting focus from $S_r(x_i)$ to $S_u(x_i)$ iteratively could maximize the effect of $Q(x_i)$ to benefit the constraint learning. Therefore, adaptive criteria weights will improve the information quantification for each unlabeled data in the framework iterations.

In this work, to dynamically control the preference of each criterion in iteration t , we introduce another hyperparameter $\epsilon \in (0, 1]$, such that $\alpha_t = \alpha_0 \epsilon^t$. The weight parameter α_t is decayed multiplicatively by ϵ in each iteration t and α_0 is predefined as an initial value of α_t . High α_0 prefer representativeness over uncertainty, and as iteration progresses, the focus of $Q(x_i)$ shifts from $S_r(x_i)$ toward $S_u(x_i)$. Low ϵ favors a rapid transition from $S_r(x_i)$ to $S_u(x_i)$. However, $S_k(\cdot)$, where $k \in \{r, u, d\}$, might be in different scales, so the weight parameter cannot be determined directly. We need to normalize each criterion as

$$\bar{S}_i(\mathbf{X}) = \frac{S_i(\mathbf{X}) - \min S_i}{\max S_i - \min S_i} \quad (11)$$

where minimum and maximum of $S_i(\mathbf{X})$ for $i \in \{u, r, d\}$ stand for the minimum and maximum over $\mathcal{U}_s$. In the

representativeness term, a higher S_r enjoys a lower representativeness, so $S_r(x_i)$ is normalized as follows:

$$\bar{S}_r(\mathbf{X}) = 1 - \frac{[S_i(\mathbf{X}) - \min S_i]}{[\max S_i - \min S_i]} \quad (12)$$

With three normalized terms, the integrated acquisition function:

$$Q(x_i) = (1 - \alpha_t - \beta)\bar{S}_u(x_i) + \alpha_t \bar{S}_r(x_i) + \beta \bar{S}_d(x_i) \quad (13)$$

The normalized measurement vector $\bar{S}_k(x_i) = \{\bar{s}_k(x_1), \bar{s}_k(x_2), \dots, \bar{s}_k(x_n)\}$, $k \in \{u, r, d\}$ contains the criteria measurements for each of $\mathcal{U}_s$ unlabeled data. Compared with the measurement vector S_u calculated upon the constraint model, S_r and S_d are extracted from the input feature of unlabeled data. In Eq. (13), the term $\bar{S}_u(x_i)$ is updated after each batch of experiments, while $\bar{S}_r(x_i)$ and $\bar{S}_d(x_i)$ are updated after each data is selected. In this sequential selection step, we adopt a heuristic way to calculate $Q(x_i)$, and the unlabeled data $x_i \in \mathcal{U}_s$ with the largest value of $Q(x_i)$ is selected in the current batch. The pseudo-code of active sampling via multi-criteria is shown in Algorithm 1.

Algorithm 1: Batch Mode Active Learning for the Unknown Feasibility Constraint

Input: Labeled set $\{\mathcal{L}, \mathcal{C}\}$, Unlabeled set $\mathcal{U}_s$, Batch size m , Weighted parameter set $\{\alpha_0, \beta\}$, Decaying parameter ϵ , $\mathbf{B}_{0,t}^{\text{Constraint}} \leftarrow \emptyset$

Output: $\mathbf{B}_{m-1,t}^{\text{Constraint}}$, selection for constraint model retraining. Ω_f predicted feasible region at the current iteration.

- 1: Train $\hat{h}(x)$ via labeled set $\{\mathcal{L}, \mathcal{C}\}$;
 - 2: Obtain prediction probability of unlabeled samples $\mathcal{U}_s$ from $\hat{h}(x)$ via Eq. (5);
 - 3: Obtain predicted feasible input (region) set $\mathcal{X}_f \in \Omega_f$ from $\hat{h}(x)$ via Eq. (6);
 - 4: Calculate $S_u(x)$ using Eq. (7) for unlabeled samples $\mathcal{U}_s$;
 - 5: **for** $i = 1, 2, \dots, m - 1$ **do**
 - 6: Calculate $S_r(x)$ using Eq. (8) for unlabeled samples $\mathcal{U}_s$;
 - 7: Calculate $S_d(x)$ using Eq. (9) for unlabeled samples $\mathcal{U}_s$;
 - 8: $x_{i,t} \leftarrow \arg\max_{x_{i,t} \in \mathcal{U}_s} \text{Eq. (13)}$;
 - 9: $\mathcal{U}_s \leftarrow \mathcal{U}_s \setminus x_{i,t}$;
 - 10: $\mathbf{B}_{i,t}^{\text{Constraint}} \leftarrow \mathbf{B}_{i-1,t}^{\text{Constraint}} \cup x_{i,t}$
 - 11: **end for**
-

3.3. Objective Optimization Loop

There are two components in the objective optimization loop (red line) in Figure 3. A surrogate model is adopted to capture the process-to-quality relationship with uncertainty quantification, which guides the subsequent unlabeled data selection for objective optimization within the estimated feasible design space. Meanwhile, to enhance the learning capability of the surrogate model with limited feasible experimental points, a pseudo-labeling technique via self-training is adopted to label infeasible designs with predicted target quality. The next sampling point is determined by an

acquisition function computed from the prediction of surrogate model training via an augmented dataset.

3.3.1. Surrogate Model

In this optimization loop, we choose GP regression to surrogate process, which captures the relationship between experimental parameters and Objective quality. Given on the same set of observed experiments $\mathbf{X}_n$, we use $\mathbf{Y} = \{y_1, y_2, \dots, y_m\}$ to represent the corresponding quality outputs of feasible experiments data $\mathbf{X}_m$. We refer to the complementary of feasible experiments as the infeasible experiments set such that $\mathbf{X}_u = \mathbf{X}_n \setminus \mathbf{X}_m$. The same as (3) in the classification problem, the objective of GP regression is to learn a specific mapping function $f(x)$, which maps an input vector to a label value, and a Gaussian prior distribution is placed over f . That is,

$$p(f|\mathbf{X}_m) \sim \mathcal{N}(f|\mu_m(x), K_m(x, x')) \quad (14)$$

where $\mu_m(x)$ is mean function and K_m is an $m \times m$ is a covariance matrix and the element of K_m is built via a kernel function $k(x, x')$. In this work, we consider the automatic relevance determination using the Matern52 kernel function, which is parameterized in terms of the kernel parameters in vector θ

$$k(x_i, x_j|\theta) = \sigma_f^2 \left(1 + \sqrt{5}r + \frac{5}{3}r^2\right) \exp(-\sqrt{5}r) \quad (15)$$

where

$$r = \sqrt{\sum_{l=1}^d \frac{(x_{il} - x_{jl})^2}{\sigma_l^2}}$$

In kernel function (15), σ_f is a non-negative overscale hyperparameter and σ_l is a different non-negative length hyperparameter for each predictor. The kernel parameter vector $\theta = [\sigma_f, \sigma_l]$ is unknown initially, and the optimal θ given on observed experiments $\mathbf{X}_n$ can be estimated by maximizing the marginal likelihood via gradient descent, which are

$$l(y_m|\mathbf{X}_m, \theta) = -\frac{1}{2}\mathbf{y}^T \mathbf{K}_m^{-1} \mathbf{y} - \frac{1}{2} \log |\mathbf{K}_m| + \frac{1}{2} \log 2\pi \quad (16)$$

Given the observed experimental data and optimal parameter vector θ , the prediction distribution of the latent function f_* for an unlabeled data x_* is

$$p(f_*|\mathbf{x}_*, \mathbf{X}_m, \mathbf{y}_m) = \mathcal{N}(\mathbb{E}(\hat{f}(x_*)), \mathbf{Var}(\hat{f}(x_*))) \quad (17)$$

where

$$\mu(x_*) = \mathbf{k}(x_*, \mathbf{X}_m) \mathbf{K}_m^{-1} \mathbf{y}_m \quad (18)$$

$$\sigma^2(x_*) = k(x_*, x_*) - \mathbf{k}(x_*, \mathbf{X}_m) \mathbf{K}_m^{-1} \mathbf{k}(x_*, \mathbf{X}_m)^\top \quad (19)$$

However, in real industrial applications, we encounter the issue of data scarcity, with only a limited number of $\mathbf{X}_m$ in the early stage of production. To enhance the surrogate model's learning ability, we incorporate a pseudo-labeling technique using a self-training mechanism (Amini *et al.*,

2022) to enlarge the training dataset for subsequent iterative selection of unlabeled data. This process involves assigning pseudo-labels obtained via

$$\widehat{\mathbf{Y}}_u = \mathbf{k}(\mathbf{X}_u, \mathbf{X}_m) \mathbf{K}_m^{-1} \mathbf{y}_m \quad (20)$$

observed infeasible samples $\mathbf{X}_u$ in each iteration. We assume the prediction of observed infeasible samples is the ground truth even though no target quality exists for observed infeasible samples. Then, we retrain the surrogate model with the augmented labeled dataset $\mathbf{X}_n$ and $\{\mathbf{Y}, \widehat{\mathbf{Y}}_u\}$ to guide our selection from in the optimization loop. Exact inference of $\widehat{\mathbf{Y}}_u$ using the GP regression model with m observed feasible data points has a time complexity of $\mathcal{O}(m^3)$ and a memory complexity $\mathcal{O}(m^2)$ (Quinero-Candela and Rasmussen, 2005). However, this computation complexity is not a concern here, as GPs are well-suited for manufacturing process design, which involves small-sized datasets.

3.3.2. Sampling Strategy in Optimization Loop

With the help of the surrogate model of the target process, an acquisition function is constructed to quantify the most informative candidate samples for objective optimization. Acquisition functions are usually derived from the $\mu(x)$ and $\sigma(x)$ generated from the surrogate model, which is easy to compute. This acquisition function allows a balance between exploitation (where the objective $\mu(x)$ is high) and exploration (where the objective $\sigma(x)$ is high). Traditionally, the sample with the maximum value of the acquisition function is selected to be labeled in the next iteration. In this study, we choose the Upper Confidence Bound (UCB) acquisition function (Srinivas *et al.*, 2009), as follows:

$$\arg \max_{x_i \in \Omega_f} \mu(x_i) + \gamma \sigma(x_i) \quad (21)$$

where γ is a hyperparameter that controls the trade-off between these two terms. The UCB function has been demonstrated as the most effective acquisition function in previous benchmarking research (Liang *et al.*, 2021). In this work, we also test the performance of our methods with different acquisition functions, including Expected Improvement (EI), Probability Improvement (PI), and Thompson Sampling (TS). Figure 3 in the Appendix depicts the convergence rate of different functions. In summary, our proposed method is compatible with different acquisition functions, but the UCB is the most effective one, especially in the case of the limited initial dataset (accounting for 5% of the whole design space). Algorithm 2 shows the details of the objective optimization.

Algorithm 2: Sampling Strategy for Objective Optimization

Input: labeled dataset $\mathcal{D}_n$ consisting of input, target quality, and feasibility $(\mathbf{x}, y_i, c_i)_{i=1}^l$, predicted feasible region space Ω_f from Algorithm 1, trade-off control parameter γ .

Output: x_*^* , Selection in the objective optimization loop
 1: **while** termination criteria are not met **do**

- 2: Train $\hat{f}(x)$ with observed feasible samples $(x_i, y_i | c_i = \text{feasible})$;
- 3: Obtain pseudo labels $\hat{y}_u$ from $\hat{f}(x)$ via Eq. (20) for observed infeasible sample $(x_i | c_i = \text{infeasible})$;
- 4: Retrain $\hat{f}(x)$ with $(x_i, y_i | c_i = \text{feasible})$ and $(x_i, \hat{y}_u | c_i = \text{infeasible})$;
- 5: Obtain prediction $\mu(x_i)$ via Eq. (18) and $\sigma(x_i)$ via Eq. (19) from $\hat{f}(x)$ for $x_i \in \Omega_f$;
- 6: $x_i^* = \operatorname{argmax}_{x_i \in \Omega_f}$ Eq. (21);
- 7: **end while**

3.4. Algorithm Summary

In this work, we introduce the **BO-ACL** framework (Bayesian Optimization with Active Constraints Learning), a novel approach that synergizes two collaborative loops for querying samples. This method is designed to optimize an objective quality of interest while simultaneously learning an unknown constraint function, all with minimal human intervention. Additionally, we employ a pool-based active learning process within a finite candidate pool predetermined by experimenters. The initial dataset $\mathcal{D}_n$ in Algorithm 3 is selected via the Latin Hypercube Sample (LHS) method. Our BO-ACL framework can be terminated when the global optimal sample is found and by expert knowledge.

Algorithm 3: Bayesian Optimization with Active Constraints Learning

Input: labeled dataset $\mathcal{D}_n$ consisting of input, target quality, and feasibility $(\mathbf{x}, y_i, c_i)_{i=1}^l$, unlabeled dataset $\mathcal{U}_s$, Weighted parameter set $\{\alpha_0, \beta, \epsilon\}$, Trade-off parameter γ , Batch size m .

Output: $\mathbf{x}_i^*$ global optimal feasible sample for target process

- 1: **while** termination criteria is not met **or** $\mathbf{x}_i^*$ is not the global optimum design **do**
- 2: Train $\hat{h}(x)$ with labeled dataset $\mathcal{D}_n$;
- 3: Obtain predicted feasible input $\mathcal{X}_f$ (feasible region Ω_f) $\in \mathcal{U}_s$ via Algorithm 1;
- 4: Obtain selection x_i^* in objective optimization loop via Algorithm 2;
- 5: Obtain active queries batch $\mathbf{B}_{m-1, n}^{\text{Constraint}}$ via Algorithm 1;
- 6: $\mathbf{X}_i^{\text{Batch}} \leftarrow \mathbf{B}_{m-1, n}^{\text{Constraint}} \cup x_i^*$;
- 7: Obtain constraint labels $\mathbf{C}_i^{\text{Batch}}$ and target quality $\mathbf{Y}_i^{\text{Batch}}$ at $\mathbf{X}_i^{\text{Batch}}$ via real experiments;
- 8: $\mathcal{D}_n \leftarrow \mathcal{D}_n \cup \{\mathbf{X}_i^{\text{Batch}}, \mathbf{Y}_i^{\text{Batch}}, \mathbf{C}_i^{\text{Batch}}\}$;
- 9: $\mathcal{U}_s \leftarrow \mathcal{U}_s \setminus \mathbf{X}_i^{\text{Batch}}$;
- 10: **end while**

Unlike traditional batch mode sampling strategies, BO-ACL implements a sequential sampling approach within each batch. This method ensures that each selection impacts subsequent choices within the same iteration. In each iteration, among the M selections, the first sample is chosen using Algorithm 2, followed by $M - 1$ samples selected

sequentially through Algorithm 1. The optimization and active learning loop influence each other's selection processes. For a detailed understanding, the pseudo-code of the BO-ACL framework is outlined in Algorithm 3.

4. Simulation Study

In this section, we evaluate the efficacy and performance of the proposed framework via two 2-D synthetic datasets used in previous research. **Simulation (1):** 2D three-hump camel function on the domain $[-2, 2] \times [-2, 2]$ (Tran *et al.*, 2019), where the function is

$$f(x) = 2x_1^2 - 1.05x_1^4 + \frac{x_1^6}{6} + x_1x_2 + x_2^2 \quad (22)$$

The global minimum of this function is $f(x^*) = 0$ at $x^* = (0, 0)$. To increase the complexity of unknown constraints, we introduce four disjoint infeasible regions, including four squares with dimensions of 1.2 for all squares. This kind of constraint setting is designed to test the capability of constraint learning in our BO-ACL method. Figure 1 in Appendix Section 1 shows the ground truth of feasibility. **Simulation (2):** 2D function as follows:

$$g(x) = x_1^2 - \sin(4x_2^2) \quad (23)$$

on the domain $[-1, 1] \times [-1, 1]$ (Basudhar *et al.*, 2012). We assume that the samples are feasible when $x_2^2 - x_1 \leq 0$. The global optimum of $g(x)$ with constraint is $f(x^*) = 1$ at $x^* = (0, 0)$. Unlike the Simulation (1) Study, the global optimum in the Simulation (2) Study is located in the feasibility boundary. Data generation is detailed in Section 1 of the Appendix. We use these two representative datasets to demonstrate their generality and applicability in different scenarios.

4.1. Efficiency of Target Objective Optimization

In this work, the BO-ACL framework presented in Section 3.4 is designed to optimize the target objective function with the least human effort following the feasibility constraint. We evaluate the convergence of our proposed BO-ACL with respect to the global optimum. The r_t are used to evaluate the convergence rate, which is defined as follows:

$$r_t = \min\{\mathcal{Y}_t\} - f(x^*) \quad (24)$$

where $\mathcal{Y}_t$ is the output of labeled feasible samples at iteration t , $f(x^*)$ is the global optimum of the target surrogate function. Our framework is compared with several baseline frameworks with the same batch size m as follows:

- Random Search (**RS**), which randomly selects m samples in each iteration;
- Constrained Bayesian Optimization (**CBO**), a BayesOpt method with constrained acquisition function for unknown constraints proposed by (Gardner *et al.*, 2014);
- **pBO-2GP-3B**, a parallel BayesOpt framework with three acquisition functions proposed by (Tran *et al.*, 2019);

In this part, we choose frameworks **CBO** and **pBO-2GP-3B** to study the effectiveness of our proposed framework. To further study the sensitivity to the initial size of labeled data, we set two different levels of initially labeled ratio: 5% and 10%. Our iterative process will be terminated when r_t equals 0, and we also set up a maximum value of process iteration t . The optimization process will be terminated when t equals 50, which means the method does not find the global optimal within 50 iterations owing to the problems we mentioned in Figure 1. For a fair comparison, our initial data are generated via the LHS method over 20 runs, and we set up the same batch size for comparison. In this part, we set up a batch size equal to 3 because there are three acquisition functions in the pBO-2GP-3B, and three is the least batch number. It is worth mentioning that there is a weighted parameter set $\{\alpha_0, \beta, \epsilon\}$ in Algorithm 1. We investigate the effect of changes in weight parameters on the results in our advanced manufacturing case study Section 5.3. In this part, we set weight parameter set $\{\alpha_0 = 0.7, \beta = 0.2, \epsilon = 0.8\}$ in Simulation (1) and weight parameter set $\{\alpha_0 = 0.6, \beta = 0.3, \epsilon = 0.7\}$ in Simulation (2) with a lower α_0 and ϵ because the overall data volume of Simulation (2) is much smaller than Simulation (1). Figure 4 depicts r_t of different frameworks versus optimization iteration number. The results demonstrate that our BO-ACL framework outperforms the baseline frameworks in the following two aspects, regardless of the initial label ratio.

Convergence speed: Our proposed framework significantly improves the average convergence speed in 20 runs in both cases. Especially in Simulation (1), the proposed method tackles the problem mentioned in Section 1. If the global optimum is located in the boundary of constraint, the **CBO** method can help us find the global optimum but wastes a lot of effort when we only have a limited number of initial data.

Performance Stability: Compared with all baseline frameworks, our proposed framework shows a more stable performance in terms of variations of r_t within 20 runs. Owing to the problem we mentioned in Figure 1, the variance of r_t in Simulation (1) will be a constant if the constraint model misclassifies the global optimum as an infeasible sample. The variance of r_t in each iteration for Simulation (2) is much smaller than in Simulation (1).

4.2. Accuracy of Constraint Estimation

In this work, learning constraint is our secondary goal, and understanding the accuracy of the feasibility constraint learned through the GP classification is crucial. These feasibility constraints, initially unknown, become progressively clearer through an iterative optimization process. The ACL loop utilizes a heuristic multi-criteria sampling strategy to facilitate learning the constraint boundary. Meanwhile, the

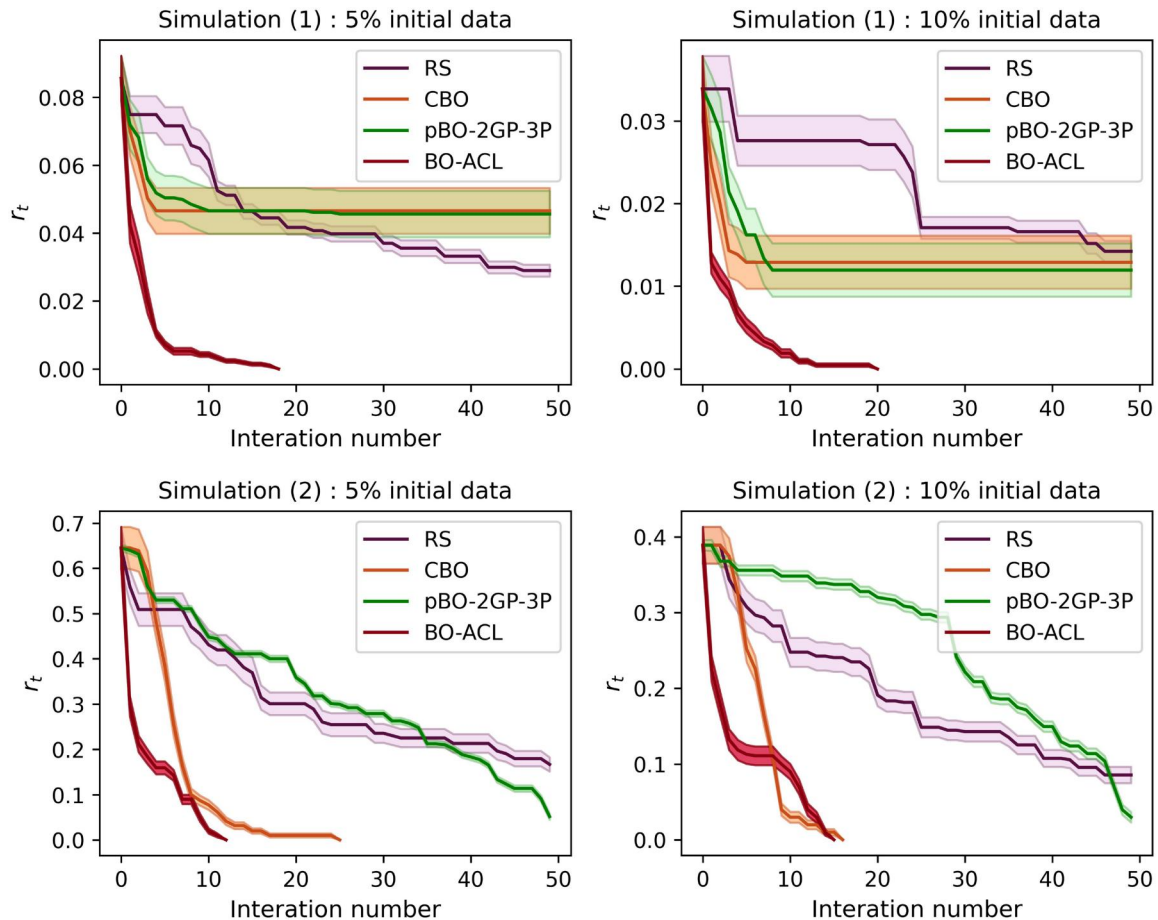


Figure 4. Convergence speed comparison for different frameworks over iterations with different sizes of initial data. Each curve represents the average results of 20 runs, and the shaded area shows a 95% confidence interval obtained from 20 repeated experiments with different sets of the initial datasets.

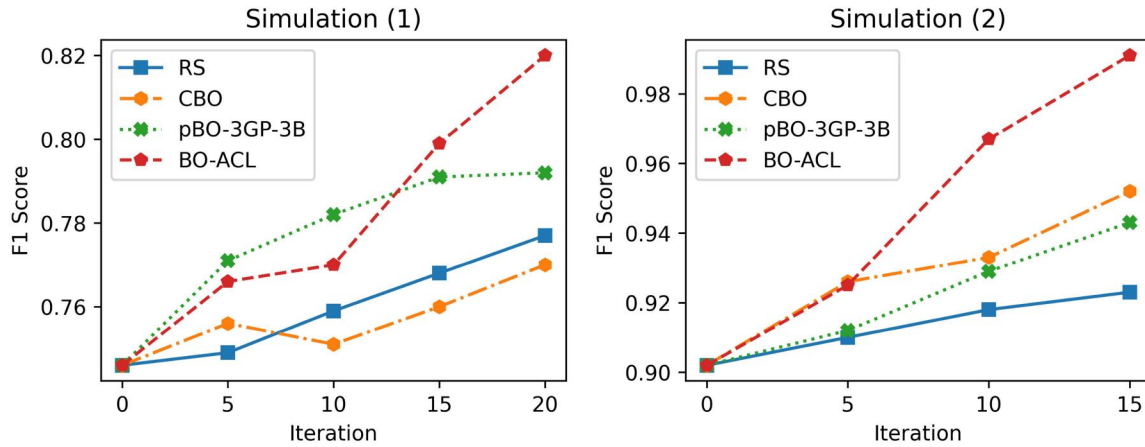


Figure 5. Accuracy comparison of constraint learning with 10% initial labeled data for different frameworks over iterations. Each curve represents the average results of 20 runs.

samples selected from the objective optimization loop also influence the performance of constraint estimation. To validate the effectiveness of our constraint estimation approach, we employ an evaluation process that are widely adopted in previous research on pool-based AL.

All labeled samples are used as training data in a pool-based AL (Wu, 2018). After each iteration, we compute the Macro F1-score as the performance measurement. The goal of pool-based AL is to create a classifier that accurately labels all candidates in the pool. However, various sampling strategies result in different labeled samples. To ensure a fair comparison, we compute the F1-score using all samples in the pool, including the D_n labeled data with their ground truth and the remaining $N - D_n$ unlabeled samples with predictions from the constraint model. Additionally, for consistent validation, each optimization framework terminates at the same iteration, specifically the average convergence iteration of the proposed framework.

Figure 5 shows the average F1-score of BO-ACL and the baseline methods for each simulation study. The results demonstrate that the proposed framework outperforms the baseline methods. However, compared with benchmark methods, our proposed method performance is worse in the first few iterations and achieves the best performance later. By contrast, as shown in Figure 4, the r_t decreases dramatically in the first few iterations. It shows that our framework is not just a simple combination of active learning and Bayesian optimization. The AL loop and optimization process are designed to work collaboratively and benefit each other from iteration to iteration. Due to page limits, we include the accuracy comparison with the 5% initial labeled data set in Appendix 2.1 to validate the proposed method's effectiveness further.

5. Case Study with Material Manufacturing Process

In this section, we implemented and validated our proposed method for the 2D material synthesis process using the chemical vapor deposition (CVD) method and demonstrated the effectiveness of our proposed BO-ACL framework. Compared with other AL-assisted optimization frameworks,

our proposed framework achieves the best performance: fast convergence to identify parameter settings with the best target quality from a small initial labeled dataset and high constraint estimation accuracy.

5.1. Data Collection and Experiment Settings

Molybdenum disulfide (MoS_2) is a typical two-dimensional transition metal dichalcogenide (2D-TMDs), enabling next-generation semiconductors and future quantum applications. Chemical vapor deposition that starts with metal-organic precursors dominates the growth studies. The material synthesis process via the CVD method requires precise adjustment of several interrelated experimental parameters to yield high-quality monolayer 2D-TMDs. Given the constraints of budget and time, material scientists aim to minimize experimental trials to efficiently discern the optimal parameter settings for successful MoS_2 growth and high-quality sample production.

In this work, a single heat source thermal CVD furnace was used to perform the experiments, and Figure 6 shows the specific layout of the CVD furnace. Following the furnace setting, five physically correlated experimental variables working together form a 5-dimensional design space with 384 points for process analysis. Appendix 3.1 describes the design space of these five experimental variables in more detail. Photoluminescence spectra (PL, excited by 473 nm laser) of MoS_2 were used to characterize the grown samples. A minimum PL intensity of 0.05 (a.u., normalized by silicon Raman peak at 521 cm^{-1}) was used to classify successful from failed growth. The fitted A-exciton linewidth σ_A (meV, obtained by Lorentzian function fits) was used to measure sample quality (Hejazi *et al.*, 2021). Recent work (Tang *et al.*, 2020) unveils the relationship between synthesis parameters and the success of experiments, which is valuable for material scientists to set up experimental conditions with a high success rate. A small value of linewidth represents a high quality of materials.

We adopted the pool-based scenario in which 384 samples in the candidate pool uniformly spread out the design space. Out of a total of 384 samples, 204 were successful

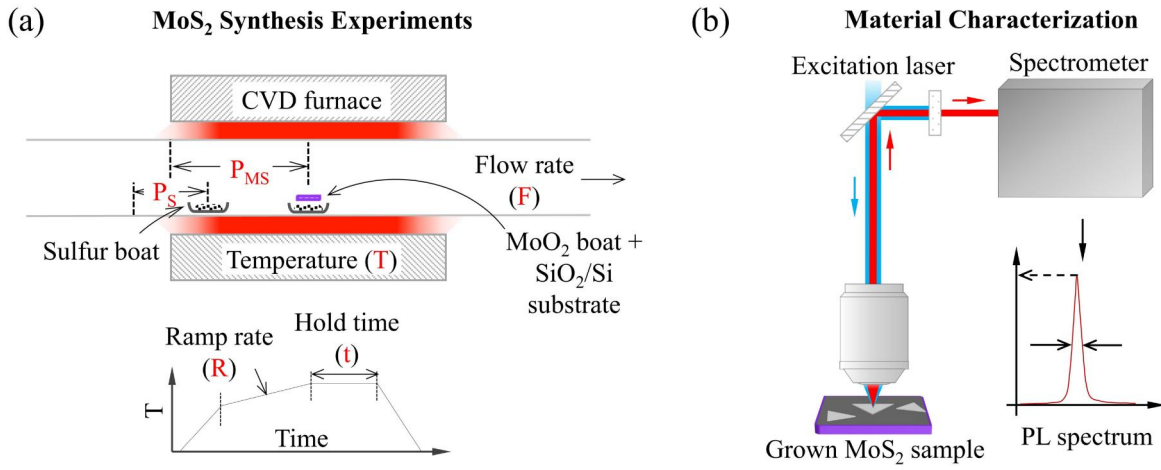


Figure 6. Schematic representation of our CVD furnace and material characterization.

design conditions, while 180 designs failed. The minimum value of the linewidth of successful samples is 0.03789 meV. For the initial design, orthogonal arrays with 48 samples generated by the Taguchi method are used as the initial dataset. Considering the variability in the initial design and the small size of the initial dataset, we chose 24 samples and 36 samples randomly from orthogonal arrays independently and replicated the experiments 20 times.

5.2. Framework Comparison

In this real-world case study, we evaluated our framework performance by comparing it with baseline frameworks: (i) **pBO-2GP-2P**, the framework (Tran *et al.*, 2019) proposed with only two acquisition functions; (ii) **UCBO**, the framework proposed (Khatamsaz *et al.*, 2023) in the previous framework with an entropy-based sampling strategy in the AL loop; (iii) **MALCBO**, the framework we proposed with a multi-criteria sampling strategy proposed in the previous framework (Kee *et al.*, 2018) considering uncertainty and diversity with a constant weighted parameter. For a fair comparison, baseline frameworks start from the same initial dataset with the proposed method, which means the r_0 is the same within all frameworks given on the same initial dataset. We adopted the same evaluation metric as shown in Eq. (24) to measure the convergence speed of each method. In our 2D material synthesis study, the theoretical minimum value of our objective A-exciton linewidth (σ_a) is around 0.004 (meV) according to the physical-based limit. However, in new manufacturing processes, such as new material discovery where optimal settings are unknown, the ground truth of experimental design can also be used to evaluate the performance. Three samples are selected in each iteration, where two queries are generated from the ACL loop, and one is generated from the objective optimization loop. The iterative optimization process will be terminated when the global optimal sample is selected. The r_i under different iterations and frameworks with different initial dataset sizes are shown in Figure 7. Our proposed framework shows superior performance over the baselines. Our proposed

framework can find the global optimal sample with an average of 20 iterations, no matter the size of the initial dataset.

For the validation of constraint estimation, we adopted the same method shown in Section 4.2. We terminated the sampling strategy at the 20th iteration, which is the average number of iterations we need to find the global optimum. Figure 8 shows the comparison of the accuracy of the constraint estimation using different methods. The proposed BO-ACL method shows the best constraint estimation results even though the estimation performance is not better than other baseline methods in the first couple of iterations. The detailed results are given in Appendix 3.3.

5.3. Sensitivity Analysis of Weight Parameters

In this section, we investigate the effect of changes in weight parameters on the optimization result. In a weighted parameter set $\{\alpha_0, \beta, \epsilon\}$, three parameters are chosen to distribute the weights assigned to each criterion in the active learning process, i.e., $\alpha_0 > 0, \beta > 0, 0 < \epsilon < 1, \alpha_0 + \beta \leq 1$. For example, when $\alpha_0 = 0$, the active learning process considers both uncertainty and diversity and by contrast, when $\alpha_0 + \beta = 1$, the active learning process considers both representativeness and diversity. Generally, the weighted parameter set is predefined by engineers considering the dimension of features and the size of the overall design space. In addition, ϵ controls the shifting rate for flavors from representativeness to uncertainty. Here, a lower value of ϵ represents a quick shift. In this sensitivity analysis, we keep the same initial dataset in Section 5.2 and set the batch size to 3. Different weight parameters are considered in this case study. We use C_m to represent the maximum number of iterations required to capture the global optimum and evaluate the convergence speed with various weighted parameters. However, in real industrial engineering applications, the global optimum of the target quality of the product is often unknown. The value of C_m is determined by the threshold of the quality measurement predefined by users based on specifications or domain expertise. However, in cases where the theoretical optimum value is known, such as in our 2D material synthesis process, the theoretical minimum value of A-exciton linewidth (σ_a) is around

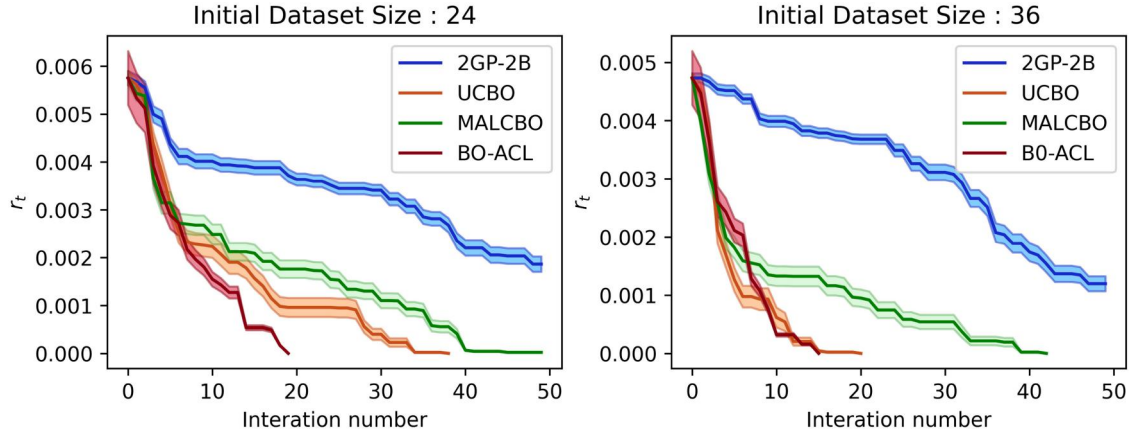


Figure 7. Convergence speed comparison of linewidth (σ_A) optimization via different frameworks over iterations.

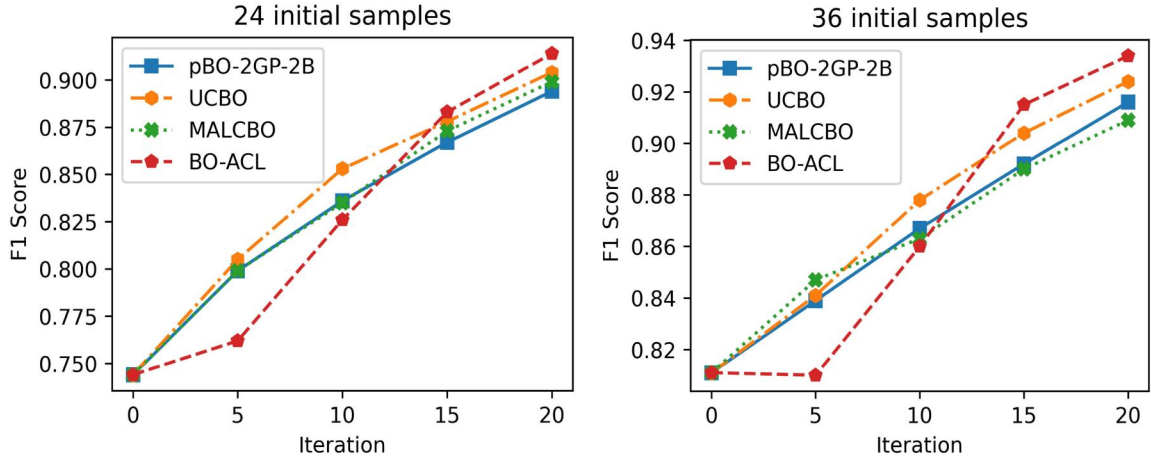


Figure 8. Accuracy comparison of constraint estimation via different methods over iterations. Each curve represents the average F1 score of 20 runs.

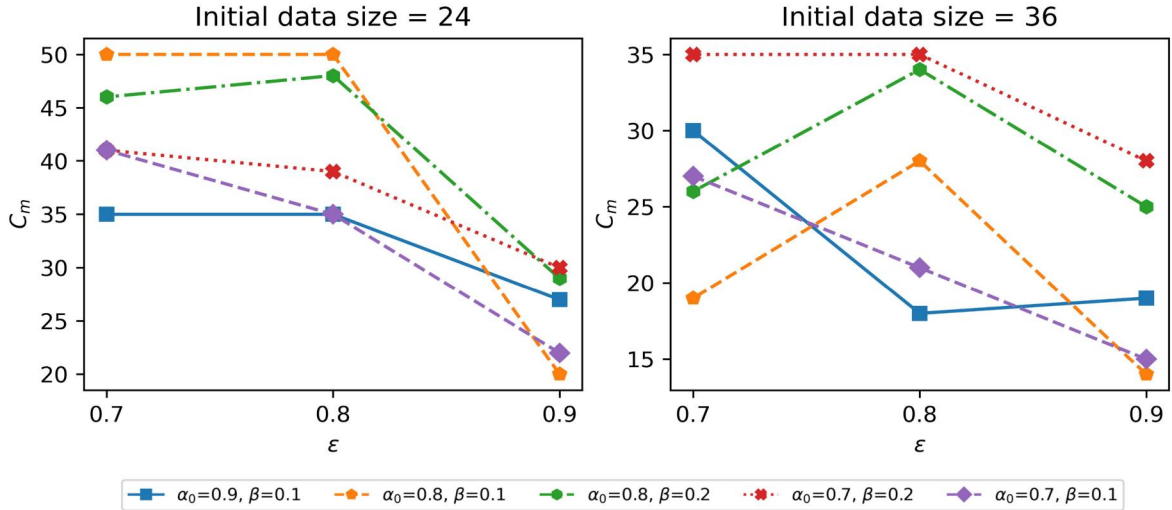


Figure 9. Convergence iteration of BO-ACL associated with different weight parameters in the case study.

0.004 (meV) according to the physics-based limit. In such instances, C_m can be determined by the inherent physical properties of the process. Figure 9 shows the performance of different weight parameters.

In this advanced material manufacturing case study, we set up three levels of $\epsilon = \{0.9, 0.8, 0.7\}$ to control the shifting focus speed from the representativeness criterion to the uncertainty.

The convergence speed of objective quality optimization is the best with $\epsilon = 0.9$, which means the focus of the representativeness term should not shift too quickly to the uncertainty term. It highlights the significance of the representativeness term we introduced in this work. Similarly, we set up three levels of $\alpha_0 = \{0.9, 0.8, 0.7\}$ to test the influence of the initial weights on the representativeness term and uncertainty term. In both cases, the higher the

Table 1. F1 score with different weight parameters

Weight Parameter	$\epsilon = 0.9$			
	24		36	
	Mean	Std	Mean	Std
$\alpha_0 = 0.8, \beta = 0.1$	0.914	0.017	0.934	0.010
$\alpha_0 = 0.9, \beta = 0.1$	0.912	0.015	0.929	0.030
$\alpha_0 = 0.7, \beta = 0.1$	0.910	0.020	0.928	0.026
$\alpha_0 = 0.8, \beta = 0.2$	0.910	0.17	0.926	0.016
$\alpha_0 = 0.8, \beta = 0.2$	0.907	0.016	0.924	0.016

value of ϵ we have, the higher our convergence speed. For practice usage of proposed methods, the general guideline for selecting hyperparameters can be summarized as follows:

- The weight parameter should be chosen based on the complexity of the dataset and the size of the initial labeled data. For a more complex dataset, a higher α_0 should be assigned. If there is less initial labeled data, a higher α_0 should be assigned. In both cases, representativeness plays a significant role at the beginning.
- The initial weight value of S_r should always be higher than S_u .
- We should set α_0 a high value, such as, $\alpha_0 \in [0.8, 1)$ and choose a high value of ϵ (close to 1) to ensure sufficient exploration of constraints while maintaining the representativeness.
- The weight assigned to S_d should be adjusted according to the value of α_0 , to ensure all three criteria are considered collectively.

While quick convergence is desirable, it does not necessarily ensure accurate constraint estimation. The weight parameters influence both the speed of convergence and the quality of constraint estimation. We follow the evaluation methodology described in Section 4.2, terminating the iterative process at the 20th iteration and calculating the F1-score to assess performance. Table 1 presents our framework's F1 scores for different parameter settings. The results indicate that the performance in constraint estimation remains relatively stable across various sets of weight parameters.

6. Conclusion

In this article, we proposed a novel parameter design framework for the constrained manufacturing process. We focused on addressing the main challenges of optimizing the process and estimating the constraint simultaneously with the least human effort. The major contribution of this study lies in that we fused AL and BayesOpt sampling in a unified framework that works collaboratively in an iterative process (i) to learn the feasibility of the experimental design and (ii) to optimize the target quality of process with minimal experiments efforts.

We first presented a generic sampling framework integrating AL and BayesOpt, in which AL performs as a feasible region estimator to help the optimization process search for the global optimum in a relatively reliable region. With a limited volume of the initial dataset, the proposed

framework can still efficiently use minimal human labeling efforts to discover the optimal experimental design. We devise a heuristic sampling strategy in the AL loop based on multiple measurements. With the multi-criteria active sampling, the search scope for the optimization process will be updated with iterations, and a better feasibility estimation will be obtained simultaneously. We also designed a dynamic weighted mechanism to distribute the focus on each measurement to improve the optimization and prediction accuracy. We verified and validated the proposed framework and algorithms using both synthetic datasets and a real-world material synthesis dataset to show superior optimization performance with minimal experimental cost. For further work, we will further investigate the framework with a human-machine collaborative optimization process with pseudo labeling in the active learning process to reduce human efforts in experiments.

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Data availability statement

The data that support the findings of this study are openly available in the supplemental material.

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